TECHNICAL PROJECT REPORT

CUSTOMER CHURN PREDICTION

An End-to-End Machine Learning Solution for Telecommunications Industry Analytics

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PROJECT REPORT

Customer Churn Prediction using Machine Learning

Abstract

Customer churn prediction is one of the most significant applications of machine learning in the telecommunications industry. Customer retention is considerably more cost-effective than acquiring new customers; therefore, identifying customers who are likely to discontinue services enables organizations to implement timely retention strategies.

This project presents a complete end-to-end machine learning pipeline for predicting customer churn using the IBM Telco Customer Churn dataset. The workflow includes data cleaning, preprocessing, exploratory data analysis, feature engineering, model development, hyperparameter tuning, and comprehensive model evaluation. Multiple classification algorithms—including Logistic Regression, Random Forest, Extra Trees, Gradient Boosting, and XGBoost—were trained and compared to identify the most suitable predictive model.

Following hyperparameter optimization using RandomizedSearchCV, the Tuned Random Forest model achieved the best overall performance with a ROC-AUC score of 0.8408, Recall of 73.39%, F1-Score of 0.6269, and an overall Accuracy of 76.87%. The model successfully identified customer tenure, total charges, monthly charges, contract type, and internet service type as the strongest predictors of customer churn.

1. Introduction

Customer churn refers to the phenomenon in which customers discontinue using a company's services. Within the telecommunications industry, customer churn significantly impacts profitability because acquiring new customers generally requires substantially higher costs than retaining existing ones.

Machine learning enables organizations to analyze historical customer information and predict future churn behavior. Such predictions allow businesses to proactively engage customers who are at high risk of leaving through personalized retention campaigns, promotional offers, and service improvements.

The primary objective of this project is to develop an accurate and reliable machine learning model capable of predicting customer churn while providing valuable business insights for customer retention.

2. Problem Statement

Telecommunication companies experience considerable financial losses due to customer attrition resulting from pricing competition, service dissatisfaction, contract flexibility, and changing customer preferences.

Without a predictive system, organizations struggle to identify high-risk customers before they leave. This project addresses this challenge by developing a supervised machine learning model capable of accurately classifying customers into churn and non-churn categories while maintaining strong recall and ROC-AUC performance.

3. Project Objectives

The primary objectives of this project are:

- Perform comprehensive data preprocessing and cleaning.
- Handle missing values and duplicate records.
- Encode categorical variables into machine-readable numerical features.
- Conduct exploratory data analysis to understand customer behavior.
- Train and evaluate multiple machine learning classification models.
- Optimize ensemble algorithms using RandomizedSearchCV.
- Compare model performance using multiple evaluation metrics.
- Identify the major business factors contributing to customer churn.
- Develop a reusable prediction model for future inference.

4. Dataset Description

The project utilizes the IBM Telco Customer Churn Dataset containing customer demographic information, subscribed services, account details, billing information, and churn status.

5. Dataset Summary

Dataset Name: IBM Telco Customer Churn Dataset
Total Records: 7,043
Original Features: 21
Target Variable: Churn

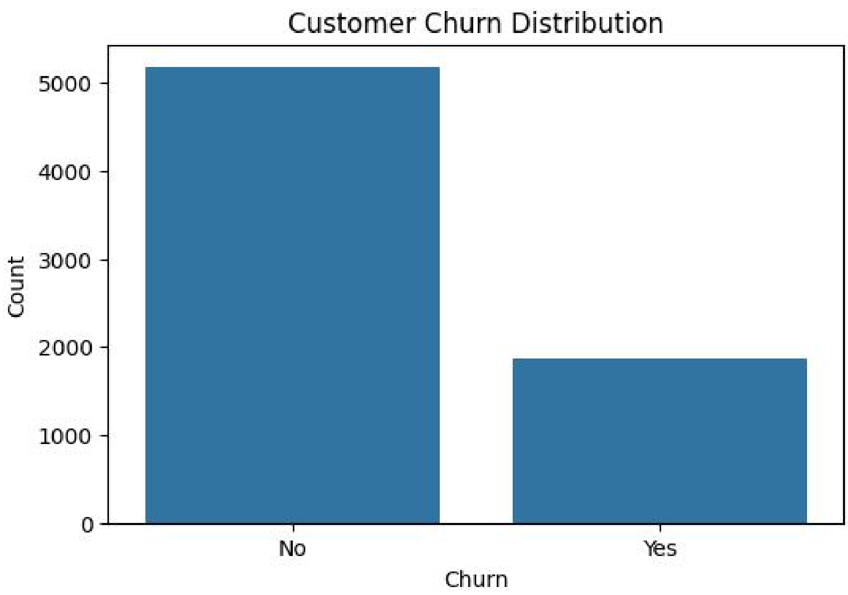
Numerical Features
- SeniorCitizen
- tenure
- MonthlyCharges
- TotalCharges

Major Categorical Features
- Gender
- Partner
- Dependents
- PhoneService
- InternetService
- OnlineSecurity
- DeviceProtection
- TechSupport
- Contract
- PaymentMethod
- PaperlessBilling

6. Exploratory Data Analysis (EDA)

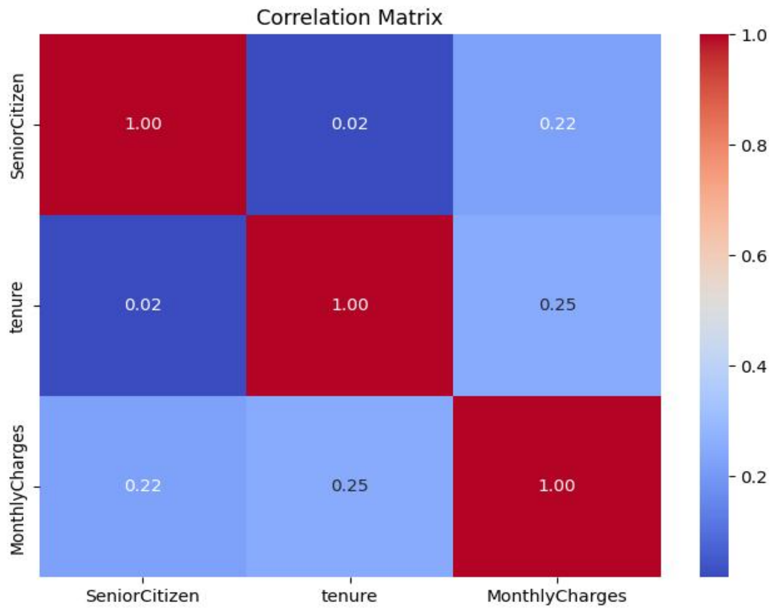
Initial analysis revealed that the dataset contains an imbalanced target variable.

Customer Distribution
- Non-Churn Customers: 5,174 (73.46%)
- Churn Customers: 1,869 (26.54%)



Correlation analysis demonstrated:

- ✧ Strong positive correlation between Tenure and TotalCharges
- ✧ Moderate positive relationship between MonthlyCharges and TotalCharges
- ✧ Weak correlation involving SeniorCitizen



EDA visualizations included:

- ✧ Customer Churn Distribution
- ✧ Correlation Heatmap
- ✧ Feature Distribution Charts

➢ Data Preprocessing

The preprocessing pipeline consisted of multiple stages designed to improve data quality and prepare the dataset for machine learning.

Data Cleaning

- Removed CustomerID
- Converted TotalCharges into numeric format
- Replaced blank values with median values
- Removed 22 duplicate records

Feature Engineering

Binary Encoding:

- Gender
- Partner
- Dependents
- PhoneService
- PaperlessBilling
- Churn

One-Hot Encoding:

- InternetService
- MultipleLines
- OnlineSecurity
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies
- Contract
- PaymentMethod

Final Dataset

- Final Samples: 7,021
- Feature Matrix: (7021 × 30)

Train-Test Split

- Training Set: 80%
- Testing Set: 20%

Feature scaling using StandardScaler was applied to numerical variables for gradient-sensitive algorithms.

7. Machine Learning Models

Five machine learning classifiers were implemented.

- Logistic Regression
- Random Forest
- Extra Trees
- Gradient Boosting
- XGBoost

Model evaluation utilized:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Baseline Model Performance

Model	Accuracy	ROC-AUC
Logistic Regression	74.09%	0.8397
Gradient Boosting	80.50%	0.8391
XGBoost	78.43%	0.8378
Random Forest	78.08%	0.8198
|Extra Trees	76.80%	0.7878

Observations

- Logistic Regression produced the highest Recall among baseline models.
- Gradient Boosting achieved the highest baseline Accuracy.
- Random Forest and Extra Trees benefited significantly from hyperparameter optimization.

Hyperparameter Optimization

RandomizedSearchCV with 5-Fold Cross Validation was applied to optimize Random Forest and Extra Trees models.

Optimized Random Forest Parameters

- n_estimators = 200
- max_depth = 10
- min_samples_split = 5
- min_samples_leaf = 4
- max_features = sqrt
- bootstrap = True

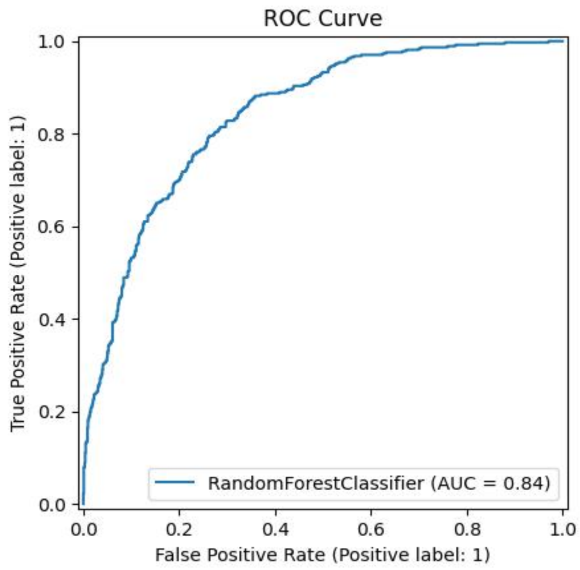
Final Model Performance

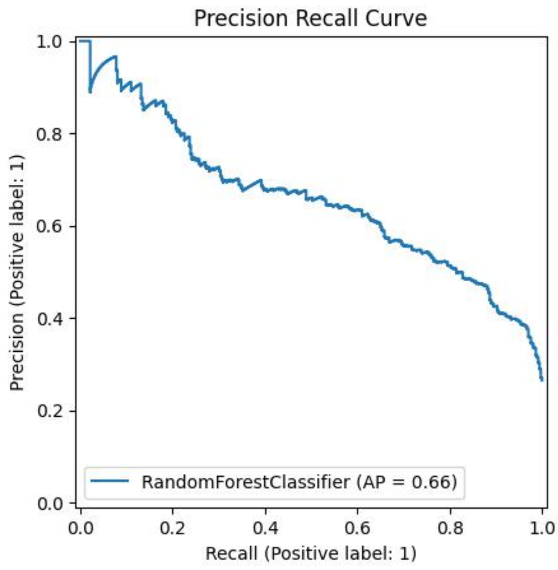
The Tuned Random Forest achieved the highest overall performance.

Metric	Value
Accuracy	76.87%
Precision	54.71%
Recall	73.39%
F1-Score	0.6269
ROC-AUC	0.8408

Cross Validation ROC-AUC:

0.8456

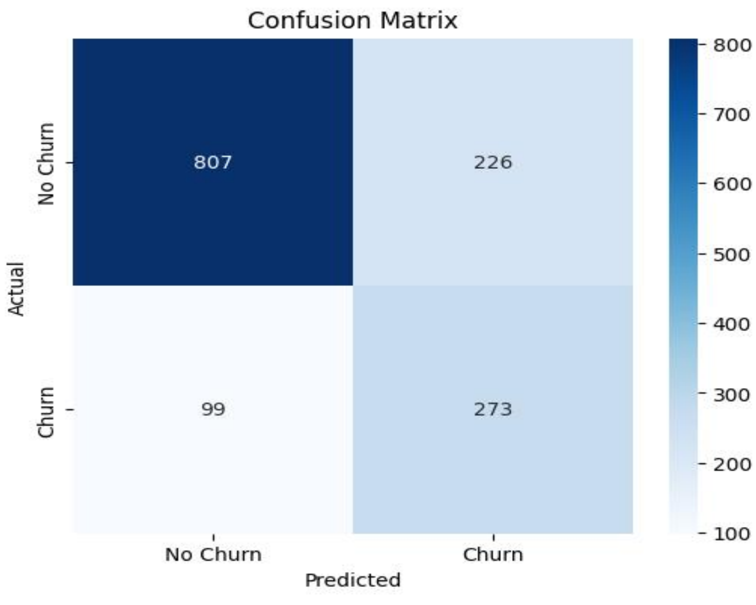




8. Confusion Matrix Analysis

The final model produced the following classification results:

- ✧ True Positives: 273
- ✧ True Negatives: 807
- ✧ False Positives: 226
- ✧ False Negatives: 99



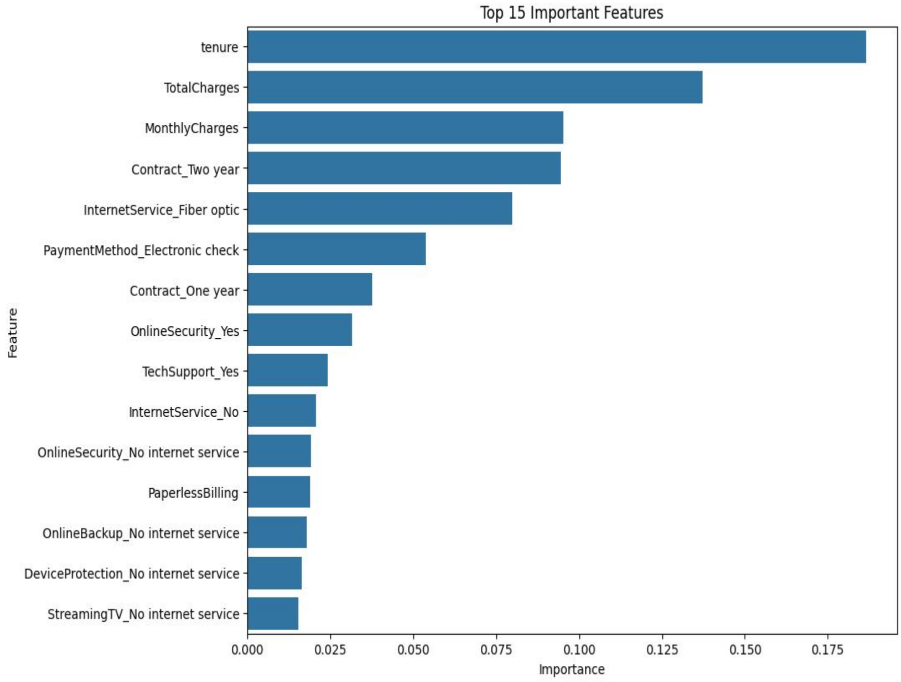
The model demonstrates strong capability in identifying customers who are likely to churn while maintaining satisfactory overall classification performance.

9. Feature Importance

The Random Forest model identified the following most influential predictors:

1. Tenure

2. TotalCharges
3. MonthlyCharges
4. Contract (Two-Year)
5. InternetService (Fiber Optic)
6. PaymentMethod (Electronic Check)
7. Contract (One-Year)
8. OnlineSecurity
9. TechSupport
10. InternetService (No)



These variables provide meaningful business insights for customer retention strategies.

Business Recommendations

Based on model findings, the following recommendations are proposed:

- ✧ Prioritize retention campaigns for customers within their first year.
- ✧ Encourage customers to adopt long-term contracts.
- ✧ Improve customer satisfaction for Fiber Optic subscribers.
- ✧ Promote automatic payment methods over electronic checks.
- ✧ Offer bundled value-added services such as Online Security and Technical Support.

Implementing these strategies can significantly reduce churn while improving long-term customer loyalty.

Deployment

The final machine learning model was exported using Joblib.

Deliverables

- ✧ Customer Churn Prediction Notebook
- ✧ customer_churn_model.pkl
- ✧ Prediction Function
- ✧ Technical Project Report

The prediction pipeline can classify new customer records into churn and non-churn categories while also generating prediction probabilities.

10. Conclusion

This project successfully developed an end-to-end machine learning solution for predicting customer churn in the telecommunications industry. Through systematic data preprocessing, feature engineering, exploratory analysis, model comparison, and hyperparameter optimization, the Tuned Random Forest classifier emerged as the most effective model with a ROC-AUC score of 0.8408.

The analysis identified customer tenure, billing patterns, contract duration, and internet service type as the strongest indicators of churn. These insights enable organizations to implement proactive retention strategies, reduce customer attrition, and improve long-term business performance.

Overall, the developed solution demonstrates the practical value of predictive analytics in supporting data-driven business decision-making and customer relationship management.

11. Future Work

Future improvements may include:

- ✧ Deep Learning models for churn prediction.
- ✧ Explainable AI (SHAP/LIME) for enhanced model interpretability.
- ✧ Real-time deployment using Flask or FastAPI.
- ✧ Automated retraining pipelines.
- ✧ Integration with cloud-based machine learning platforms.

12. References

1. IBM Sample Data Sets – Telco Customer Churn Dataset.
2. Scikit-learn Documentation.
3. XGBoost Official Documentation.
4. Pandas Documentation.
5. NumPy Documentation.
6. Matplotlib Documentation.
7. Seaborn Documentation.
8. Joblib Documentation.